



DCB × ALSHUMOOKH — Integration Roadmap

Ledger-to-Ledger ISO 20022 API Settlement — Step-by-Step Implementation Plan

ESTIMATED TIMELINE

Phase 1 — Legal
Weeks 1–3

Phase 2 — Technical
Weeks 4–9

Phase 3 — Security
Weeks 10–12

Phase 4 — Go Live
Weeks 13–14

PHASE 1 — LEGAL & COMPLIANCE PREREQUISITES

- | | | |
|----------|---|-----------------|
| 1 | Verify DCB Banking License
Request official proof of licensing from CBUAE (Central Bank of UAE) or DFSA. Confirm DCB is authorized to transmit and hold USD cross-border. | CRITICAL |
| 2 | Sign Correspondent Banking Agreement
Execute a formal legal agreement defining: liability for failed transfers, settlement finality, dispute resolution, and obligations of each party. | CRITICAL |
| 3 | AML / KYC Due Diligence on DCB
Conduct full KYC on Digital Commercial Bank Ltd and its principals. Verify beneficial ownership and source of funds policy. | CRITICAL |
| 4 | Define Settlement Finality Rules
Agree legally on what constitutes a completed transfer: Is receipt of pacs.008 message sufficient, or must funds clear a Nostro account first? | REQUIRED |

■ **Do NOT begin any technical development until all Phase 1 items are signed and verified.**



PHASE 2 — TECHNICAL DEVELOPMENT

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Build ISO 20022 Receiver Endpoint (pacs.008)
Add a new FastAPI endpoint: POST /api/v1/iso20022/inbound — accepts XML pacs.008 messages, parses BIC, IBAN, amount, currency, and creditor reference fields.

DEVELOP
- 6

Implement OAuth2 Token Authentication
Add OAuth2 client credentials flow. DCB authenticates and receives a Bearer token before every API call. Token expiry and refresh logic must be implemented.

DEVELOP
- 7

Configure mTLS (Mutual TLS) Certificate Exchange
Generate server certificate for api.alshumookh-pay.com. Exchange client certificate CN=fluxliqdaes.cloud with DCB. Configure Nginx/reverse proxy for mTLS enforcement.

DEVELOP
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Build XML Parser & Validation Layer
Parse incoming pacs.008 XML. Validate schema against ISO 20022 XSD. Extract: amount, currency, sender BIC, creditor account, end-to-end ID, and settlement date.

DEVELOP
- 9

Build Settlement Reconciliation Engine
Map received pacs.008 message to a client account in ALSHUMOOKH database. Credit the account balance, generate internal settlement record, and trigger notifications.

DEVELOP
- 10

Send ISO 20022 Acknowledgement (pacs.002)
On successful receipt, return a pacs.002 XML confirmation to DCB confirming settlement acceptance with status ACSC (Accepted Settlement Completed).

DEVELOP

✓ Estimated development time: 5–6 weeks with 1 senior backend developer.



PHASE 3 — SECURITY & TESTING

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IP Whitelisting — DCB Servers Only
Restrict ISO 20022 inbound endpoint to DCB's IP range only (185.229.57.76 confirmed). All other IPs return 403. Configure at Nginx and application layer.

SECURITY
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Payload Signature Verification (HMAC / AES-256)
Verify every incoming pacs.008 message carries a valid HMAC-SHA256 signature using a shared secret key agreed with DCB. Reject any unsigned payload immediately.

SECURITY
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End-to-End UAT Testing with DCB
Run full User Acceptance Testing: send 1 USD test, verify XML parsing, check credit to client account, confirm pacs.002 response is received by DCB.

TESTING
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Audit Trail & Compliance Logging
Log every inbound ISO 20022 message with: timestamp, sender BIC, amount, currency, XML raw payload, and settlement status. Retain for 7 years minimum.

COMPLIANCE

PHASE 4 — GO LIVE & MONITORING

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Production Deployment & DNS / SSL Validation
Deploy ISO 20022 endpoint to production on api.alshumookh-pay.com. Validate SSL chain, mTLS handshake, and OAuth2 token flow in live environment.

GO LIVE
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Real-Time Settlement Dashboard
Add ISO 20022 inbound transfers section to the Admin Dashboard showing: sender, amount, currency, XML reference, status, and settlement confirmation.

GO LIVE

INTEGRATION SUMMARY

Total Phases:	4 Phases
Total Steps:	16 Steps
Estimated Duration:	13–14 Weeks
Key Standard:	ISO 20022 (pacs.008 / pacs.002)
Authentication:	mTLS + OAuth2 Bearer Token
Critical Prerequisite:	DCB Banking License + Legal Agreement

■ The legal phase (Phase 1) is non-negotiable — technical work must not start before it is complete.